

B. Tech 3rd Semester 1st Sessional Examination 2024
Sub: Mathematics for Civil Engineering (BMA23211T)
Branch- CE, Total Marks-30, Date: 09/09/2024

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1. Solve (any one):

(a) $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} + u; u(x, 0) = 6e^{-3x}$

(b) $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}; u(0, y) = 8e^{-3y}$

2. Answer any one of the following:

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(a) Solve the one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ subject to boundary conditions $u(0, x) = 0 = u(l, t)$ and initial condition $u(x, 0) = u_0$

(b) A tightly stretched string of length l is initially in a position given by $y = y_0 \sin \frac{\pi x}{l}$. If it is released from rest from this position, find the displacement $y(x, t)$

3. Find the Laplace Transform of (any three)

5X3=15

(a) $\sin 4t \cos 5t$

(b) $t^2 e^{-2t}$

(c) $t e^{at} \sin at$

(d) $e^{-3t} (2 \cos 5t - 3 \sin 5t)$